How to Scale Your Model: Part 1 Roofline Exercises

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1 int8 Matmul

We consider the operation

$$\text{int8}[B, D] \cdot_D \text{int8}[D, F] \;\rightarrow\; \text{int8}[B, F],$$

where $A \cdot_D B$ denotes contraction (matrix multiplication) over the shared index D. An int8 value occupies 1 byte and a bf16 value occupies 2 bytes; all calculations assume each tensor is read from HBM once and the output written back once.

Since int8 uses 1 byte per value, the bytes loaded are $BD + DF$ and the bytes written back are BF, giving total memory traffic $BD + DF + BF$. There are BF output entries, each a dot product over D requiring D multiplications and $D - 1$ additions, so the exact operation count is $BF(2D - 1) \approx 2BDF$. The arithmetic intensity is therefore

$$I = \frac{2BDF}{BD + DF + BF} \approx \frac{2BDF}{DF} = 2B, \tag{1}$$

where the approximation assumes B is small relative to D and F, so the DF term dominates the denominator. The critical arithmetic intensity is

$$I_{\text{crit}} = \frac{3.94 \times 10^{14}}{8.2 \times 10^{11}} \approx 480, \tag{2}$$

so the operation becomes compute-bound when $2B \gtrsim 480$, i.e. $B \gtrsim 240$. The runtime bounds follow from

$$T_{\text{math}} = \frac{2BDF}{3.94 \times 10^{14}}, \qquad T_{\text{comms}} = \frac{BD + DF + BF}{8.2 \times 10^{11}}, \tag{3}$$

with a reasonable lower bound $T_{\text{lower}} = \max(T_{\text{math}}, T_{\text{comms}})$ and upper bound $T_{\text{upper}} = T_{\text{math}} + T_{\text{comms}}$.

2 int8 Weights and bf16 Activations

We consider the operation

$$\text{bf16}[B, D] \cdot_D \text{int8}[D, F] \;\rightarrow\; \text{bf16}[B, F].$$

The operation count is still $\approx 2BDF$. The bf16 activations cost $2BD$ bytes to load, the int8 weights cost DF bytes, and the bf16 output costs $2BF$ bytes, for total traffic $2BD + DF + 2BF$. Hence

$$I = \frac{2BDF}{2BD + DF + 2BF} \approx \frac{2BDF}{DF} = 2B. \tag{4}$$

The critical arithmetic intensity for bf16 compute is

$$I_{\text{crit}} = \frac{1.97 \times 10^{14}}{8.2 \times 10^{11}} \approx 240, \tag{5}$$

so the operation becomes compute-bound when $2B \gtrsim 240$, i.e. $B \gtrsim 120$.

3 Roofline Plot

The roofline plot uses the exact memory traffic from Section 2, $2BD + DF + 2BF$, rather than the large-D, F approximation. For each B, the achieved throughput is

$$\text{Achieved FLOPs/s} = \frac{2BDF}{\max(T_{\text{math}}, T_{\text{comms}})},$$

where

$$T_{\text{math}} = \frac{2BDF}{1.97 \times 10^{14}}, \qquad T_{\text{comms}} = \frac{2BD + DF + 2BF}{8.2 \times 10^{11}}. \tag{6}$$

The two plotted cases are $D = F = 4096$ and $D = F = 1024$.

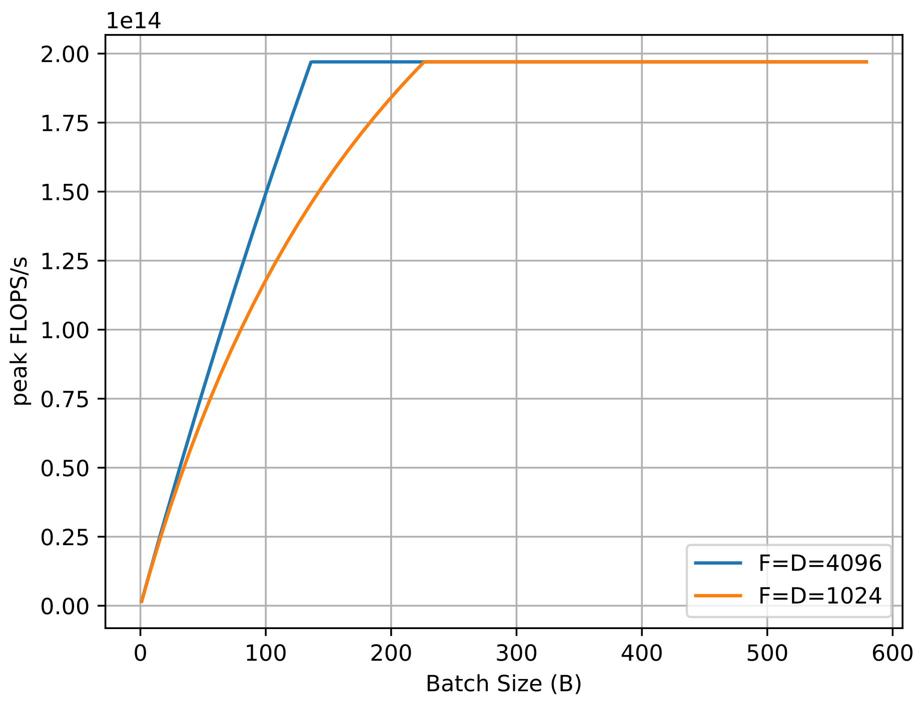

Figure 1: Achieved throughput versus batch size for $D = F = 4096$ and $D = F = 1024$.

4 Per-Batch int8 Matrices

We consider the operation

$$\text{int8}[B, D] \cdot_D \text{int8}[B, D, F] \ \to \ \text{int8}[B, F].$$

There are still BF output entries, each a dot product over D, so the operation count is $\approx 2BDF$. Memory traffic is larger because each batch element carries its own $D \times F$ matrix: loading the first input costs BD bytes, the second costs BDF bytes, and the output costs BF bytes, for total traffic $BD + BDF + BF$. Therefore

$$I = \frac{2BDF}{BD + BDF + BF} = \frac{2DF}{D + DF + F} \approx \frac{2DF}{DF} = 2, \tag{7}$$

where factoring out B shows the intensity is independent of batch size, and the DF term dominates for large D, F. Unlike ordinary matrix multiplication, increasing the batch size does not improve arithmetic intensity, because each batch element has its own matrix. This operation is therefore generally memory-bandwidth bound.

5 H100 SXM bf16 Matmul

For the H100 SXM, the reported bf16 Tensor Core performance is 1.979×10^{15} FLOPs/s, but this figure assumes structured sparsity. The dense bf16 value is half of this:

$$\frac{1.979 \times 10^{15}}{2} = 9.89 \times 10^{14} \text{ FLOPs/s}.$$

With HBM bandwidth $3.35\,\text{TB/s} = 3.35 \times 10^{12}$ bytes/s, the critical arithmetic intensity is

$$I_{\text{crit}} = \frac{9.89 \times 10^{14}}{3.35 \times 10^{12}} \approx 295. \tag{8}$$

For a standard bf16 matmul the arithmetic intensity is $I \approx B$, so the operation becomes compute-bound when $B \gtrsim 295$.